

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA0713
COURSE NAME: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

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Code of Conduct:

*I **Shane Rooney**[192511057] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Shane Rooney.A

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Question : 1

A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyse the missing layers and explain how their functions are essential for reliable communication.

Answer : 1

Missing OSI Layers

*Application → Presentation → Session → Transport → Network → Data Link → Physical.
Session Layer: establishes, manages and terminates sessions. Data Link Layer: framing, MAC addressing, error detection and reliable node-to-node delivery.*

Question : 2

Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Answer : 2

Data Units

Application/Presentation/Session: Data

Transport: Segment

Network: Packet

Data Link: Frame

Physical: Bits

Each layer encapsulates data by adding its header, enabling reliable transmission.

Question : 3

Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Answer :3

MAC vs IP

MAC addresses provide local delivery within a LAN. IP addresses provide global routing between networks. Together they ensure end-to-end communication.

Question : 4

A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Answer : 4

Error Control

Data Link Layer detects transmission errors using CRC/checksum and retransmits frames if needed. Transport Layer provides end-to-end reliability using acknowledgements, sequencing and retransmission, ensuring accurate delivery.

Question : 5

Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Answer : 5

Web Browsing

Application(HTTP/HTTPS) → Presentation(Encryption/TLS) → Session(Connection) → Transport(TCP) → Network(IP) → Data Link(Frame/MAC) → Physical(Bits).

Failure in any layer interrupts higher-layer communication, so web pages cannot load correctly.

Question : 6

A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

Answer : 6

Troubleshooting

Physical and Data Link are working, but the Network layer has failed. Since IP routing fails, Transport and Application layers cannot communicate. The concept map helps identify the failed layer quickly, reducing troubleshooting time.